CSC 370 PaperTrade, Minimal Basis and FD Projection

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Overview

This document does two things for the FDs of the `stock_trading` database:

1. derives a **minimal basis** for the full FD set F of the universal relation U;
2. **projects** F onto every relation produced by the BCNF decomposition in the *Lossless Join Proof*: the two intermediate relations U_1 and U_2, and the four final relations `user`, `stock`, `holds` and `transaction`. Each projection is then reduced to a minimal basis.

The projections show that every BCNF violation used in the decomposition really holds in the relation where it was used, and that the four final relations together preserve every FD of F.

Notation

As in the *Lossless Join Proof*, U holds every attribute of the database, with the two quantity attributes renamed t_quantity (shares in a trade) and h_quantity (shares currently held). We group attributes that always appear together:

$$N = \{\text{name}, \text{email}\} \qquad S = \{\text{company_name}, \text{opening_price}, \text{closing_price}, \text{exchange}\}$$
$$T = \{\text{transaction_type}, \text{t_quantity}, \text{price}, \text{time}\} \qquad P = \{\text{h_quantity}, \text{average_price}\}$$

$$U = \{\text{transaction_id}, \text{user_id}, \text{ticker}\} \cup N \cup S \cup T \cup P$$

The FD set F, from the *Functional Dependencies and Candidate Keys* document:

- FD_1: user_id $\to$ name, email
- FD_2: email $\to$ user_id
- FD_3: ticker $\to$ company_name, opening_price, closing_price, exchange
- FD_4: transaction_id $\to$ user_id, ticker, transaction_type, t_quantity, price, time
- FD_5: user_id, ticker $\to$ h_quantity, average_price

1 Minimal Basis of F

A set of FDs G is a **minimal basis** for F if G is equivalent to F ($G^+ = F^+$) and:

(1) every right-hand side is a single attribute;

(2) no FD can be removed, because no $X \to A$ in G follows from the other FDs of G;

(3) no left-hand side can be reduced, because for no $X \to A$ in G and $B \in X$ does $X \setminus \{B\} \to A$ follow from G.

We build G in three steps, one for each condition.

Step 1: Single-Attribute Right-Hand Sides

By decomposition, each FD $X \to A_1 \cdots A_k$ is equivalent to the k FDs $X \to A_1, \ldots, X \to A_k$ (and union recombines them). Splitting FD_1 to FD_5 gives 15 FDs:

From	Label	FD
FD_1	g_1, g_2	user_id $\to$ name, user_id $\to$ email
FD_2	g_3	email $\to$ user_id
FD_3	g_4 – g_7	ticker $\to$ company_name, ticker $\to$ opening_price, ticker $\to$ closing_price, ticker $\to$ exchange
FD_4	g_8 – g_{13}	transaction_id $\to A$ for each $A \in$ {user_id, ticker, transaction_type, t_quantity, price, time}
FD_5	g_{14}, g_{15}	user_id, ticker $\to$ h_quantity, user_id, ticker $\to$ average_price

Step 2: Reduce Left-Hand Sides

Only g_{14} and g_{15} have more than one attribute on the left. For each, we test whether either attribute alone already determines the right-hand side, using closures under the full set G:

$$\{\text{user_id}\}^+ = \{\text{user_id}, \text{name}, \text{email}\} \qquad (g_1, g_2, \text{ then } g_3 \text{ adds nothing new})$$
$$\{\text{ticker}\}^+ = \{\text{ticker}, \text{company_name}, \text{opening_price}, \text{closing_price}, \text{exchange}\} \qquad (g_4 \text{ to } g_7)$$

Neither closure contains h_quantity or average_price. So neither attribute can be dropped from g_{14} or g_{15}: a position is determined by the user and the stock together, not by either one.

Step 3: Remove Redundant FDs

An FD $X \to A$ is redundant if $A \in X^+$, where the closure is computed with $X \to A$ itself removed.

Shortcut. If $A \notin X$ and $X \to A$ is the *only* FD with A on its right-hand side, it cannot be redundant. Without it, the closure algorithm has no FD that adds A, and A is not in the starting set X.

In G, every attribute is on the right-hand side of exactly one FD except user_id, which is on the right of g_3 (email $\to$ user_id) and g_8 (transaction_id $\to$ user_id). By the shortcut, the other 13 FDs are not redundant. We test the remaining two directly:

- g_3 (email $\to$ user_id): without g_3, no FD has email alone on its left, so $\{\text{email}\}^+ = \{\text{email}\}$. This does not contain user_id, so g_3 is not redundant.
- g_8 (transaction_id $\to$ user_id): without g_8,

$$\{\text{transaction_id}\}^+ = \{\text{transaction_id}, \text{ticker}\} \cup T \cup S \quad (g_9 \text{ to } g_{13}, \text{ then } g_4 \text{ to } g_7)$$

Reaching user_id would take g_3, which needs email, which only g_2 gives, which needs user_id. The closure never gets there, so g_8 is not redundant.

Result

No FD can be removed and no left-hand side can be reduced, so all 15 FDs of Step 1 form a minimal basis G of F.

Minimal bases are not unique. g_1 (user_id $\to$ name) could be replaced by email $\to$ name, and the set would still be minimal and equivalent to F, since user_id $\to$ email $\to$ name recovers g_1 by transitivity. We keep g_1 because it matches how `user` is keyed.

2 Projecting FDs

The Projection Algorithm

The projection of F onto a relation $R_i \subseteq U$ is the set of all FDs $X \to A$ in F^+ with $X \cup \{A\} \subseteq R_i$. It is computed by taking every $X \subseteq R_i$, finding the closure X^+ under F in all of U, and adding

$$X \to (X^+ \cap R_i) \setminus X$$

The closure must be taken in U, not in R_i, because a derivation can pass through attributes outside R_i. The result is then reduced to a minimal basis.

Trying all $2^{|R_i|}$ subsets is unnecessary. Three rules let us skip subsets without missing any FD:

(R1) **Empty set:** no FD has an empty left-hand side, so $\emptyset^+ = \emptyset$ gives nothing.

(R2) **Supersets of a superkey:** if $X^+ \supseteq R_i$, every $Y \supseteq X$ only gives FDs that follow from $X \to R_i$ by augmentation and decomposition, so they can be skipped.

(R3) **Attributes on no left-hand side:** let $L = \{\text{transaction_id}, \text{user_id}, \text{email}, \text{ticker}\}$, the attributes that appear on some left-hand side of F. If $X = Y \cup Z$ with $Z \cap L = \emptyset$, then adding Z cannot fire any new FD, so $X^+ = Y^+ \cup Z$. The FD from X is then an augmentation of the FD from Y. So only subsets of $R_i \cap L$ need to be tried.

Closures Used Below

These are computed once, under F in all of U:

$$\begin{aligned}
\{\text{user_id}\}^+ &= \{\text{user_id}\} \cup N && (FD_1)\\
\{\text{email}\}^+ &= \{\text{user_id}\} \cup N && (FD_2,\ \text{then } FD_1)\\
\{\text{ticker}\}^+ &= \{\text{ticker}\} \cup S && (FD_3)\\
\{\text{user_id}, \text{ticker}\}^+ &= \{\text{user_id}, \text{ticker}\} \cup N \cup S \cup P && (FD_1, FD_3, FD_5)\\
\{\text{transaction_id}\}^+ &= U && (FD_4,\ \text{then } FD_1, FD_3, FD_5)
\end{aligned}$$

email is in no relation except `user`, so it only matters there.

2.1 Intermediate Relation U_1

$U_1 = \{\text{transaction_id}, \text{user_id}, \text{ticker}\} \cup S \cup T \cup P$, left after Step 1 of the decomposition removes N. Here $R_i \cap L = \{\text{transaction_id}, \text{user_id}, \text{ticker}\}$.

X	$(X^+ \cap U_1) \setminus X$	**Projected FD**
{transaction_id}	{user_id, ticker} $\cup S \cup T \cup P$	transaction_id $\to$ user_id, ticker, S, T, P
{user_id}	$\emptyset$ (N is not in U_1)	none
{ticker}	S	ticker $\to S$
{user_id, ticker}	$S \cup P$	user_id, ticker $\to S, P$

Every other subset of $R_i \cap L$ contains transaction_id, so it is skipped by (R2).

Minimal basis of the projection. After splitting right-hand sides:

- transaction_id $\to A$ for $A \in S$ is redundant: transaction_id $\to$ ticker $\to A$.
- transaction_id $\to A$ for $A \in P$ is redundant: transaction_id $\to$ user_id, ticker $\to A$.
- user_id, ticker $\to A$ for $A \in S$ reduces on the left to ticker $\to A$, which is already in the set.

$$F_{U_1} = \{\text{ticker} \to A : A \in S\} \cup \{\text{user_id}, \text{ticker} \to A : A \in P\} \cup \{\text{transaction_id} \to A : A \in \{\text{user_id}, \text{ticker}\} \cup T\}$$

That is FD_3, FD_5 and FD_4, split. FD_1 and FD_2 do not project, because $N \cap U_1 = \emptyset$. Since $\{\text{ticker}\}^+ \cap U_1 = \{\text{ticker}\} \cup S \neq U_1$, the FD ticker $\to S$ violates BCNF in U_1, which is the violation Step 2 of the decomposition removes.

2.2 Intermediate Relation U_2

$U_2 = \{\text{transaction_id}, \text{user_id}, \text{ticker}\} \cup T \cup P$, left after Step 2 removes S. Again $R_i \cap L = \{\text{transaction_id}, \text{user_id}, \text{ticker}\}$.

X	$(X^+ \cap U_2) \setminus X$	**Projected FD**
{transaction_id}	{user_id, ticker} $\cup T \cup P$	transaction_id $\to$ user_id, ticker, T, P
{user_id}	$\emptyset$	none
{ticker}	$\emptyset$ (S is not in U_2)	none
{user_id, ticker}	P	user_id, ticker $\to P$

A derived FD appears. transaction_id $\to$ h_quantity is not in F. It holds in U_2 because FD_4 and FD_5 chain through user_id, ticker by transitivity. Projection has to keep FDs like this: computing closures in U, rather than only copying FDs from F whose attributes fit in U_2, is what finds them. In this case it is redundant once FD_4 and FD_5 are both present, so the minimal basis drops it:

$$F_{U_2} = \{\text{user_id}, \text{ticker} \to A : A \in P\} \cup \{\text{transaction_id} \to A : A \in \{\text{user_id}, \text{ticker}\} \cup T\}$$

Since $\{\text{user_id}, \text{ticker}\}^+ \cap U_2 = \{\text{user_id}, \text{ticker}\} \cup P$ does not contain transaction_id, {user_id, ticker} is not a superkey of U_2. FD_5 violates BCNF here, which is the violation Step 3 removes.

2.3 Final Relations

$\texttt{user} = \{\textbf{user_id}, \textbf{name}, \textbf{email}\}$

$R_i \cap L = \{\text{user_id}, \text{email}\}$. $\{\text{user_id}, \text{email}\}$ is a superset of a superkey, so it is skipped by (R2).

X	$(X^+ \cap \texttt{user}) \setminus X$	**Projected FD**
$\{\text{user_id}\}$	$\{\text{name}, \text{email}\}$	$\text{user_id} \to \text{name}, \text{email}$
$\{\text{email}\}$	$\{\text{user_id}, \text{name}\}$	$\text{email} \to \text{user_id}, \text{name}$

Splitting gives $\text{user_id} \to \text{name}$, $\text{user_id} \to \text{email}$, $\text{email} \to \text{user_id}$ and $\text{email} \to \text{name}$. Testing $\text{email} \to \text{name}$ first: without it, $\{\text{email}\}^+ = \{\text{email}, \text{user_id}, \text{name}\}$, which contains name, so it is redundant. The other three each have the only FD for their right-hand side, so none can be removed:

$$F_{\texttt{user}} = \{\text{user_id} \to \text{name},\ \text{user_id} \to \text{email},\ \text{email} \to \text{user_id}\} = \{g_1, g_2, g_3\}$$

$\texttt{stock} = \{\textbf{ticker}\} \cup S$

$R_i \cap L = \{\text{ticker}\}$, and $\{\text{ticker}\}^+ \cap \texttt{stock} = \texttt{stock}$:

$$F_{\texttt{stock}} = \{\text{ticker} \to A : A \in S\} = \{g_4, \ldots, g_7\}$$

$\texttt{holds} = \{\textbf{user_id}, \textbf{ticker}\} \cup P$

$R_i \cap L = \{\text{user_id}, \text{ticker}\}$.

X	$(X^+ \cap \texttt{holds}) \setminus X$	**Projected FD**
$\{\text{user_id}\}$	$\emptyset$ (N is not in `holds`)	none
$\{\text{ticker}\}$	$\emptyset$ (S is not in `holds`)	none
$\{\text{user_id}, \text{ticker}\}$	P	$\text{user_id}, \text{ticker} \to P$

Neither attribute alone gives anything, so the left-hand side cannot be reduced:

$$F_{\texttt{holds}} = \{\text{user_id}, \text{ticker} \to \text{h_quantity},\ \text{user_id}, \text{ticker} \to \text{average_price}\} = \{g_{14}, g_{15}\}$$

$\texttt{transaction} = \{\textbf{transaction_id}, \textbf{user_id}, \textbf{ticker}\} \cup T$

$R_i \cap L = \{\text{transaction_id}, \text{user_id}, \text{ticker}\}$.

X	$(X^+ \cap \texttt{transaction}) \setminus X$	**Projected FD**
$\{\text{transaction_id}\}$	$\{\text{user_id}, \text{ticker}\} \cup T$	$\text{transaction_id} \to \text{user_id}, \text{ticker}, T$
$\{\text{user_id}\}$	$\emptyset$	none
$\{\text{ticker}\}$	$\emptyset$	none
$\{\text{user_id}, \text{ticker}\}$	$\emptyset$ (N, S and P are not in `transaction`)	none

$\{\text{user_id}, \text{ticker}\}$ determines nothing here. FD_5 does not carry over, because its right-hand side P is in `holds`, not `transaction`. This is why one user can trade the same stock many times:

$$F_{\texttt{transaction}} = \{\text{transaction_id} \to A : A \in \{\text{user_id}, \text{ticker}\} \cup T\} = \{g_8, \ldots, g_{13}\}$$

3 Dependency Preservation

The minimal bases of the four final projections are disjoint, and together they are exactly G:

$$F_{\texttt{user}} \cup F_{\texttt{stock}} \cup F_{\texttt{holds}} \cup F_{\texttt{transaction}} = \{g_1, g_2, g_3\} \cup \{g_4, \ldots, g_7\} \cup \{g_{14}, g_{15}\} \cup \{g_8, \ldots, g_{13}\} = G$$

G is a minimal basis of F, so $G^+ = F^+$. Every FD of F can therefore be enforced inside a single relation, by that relation's primary key or the `UNIQUE` constraint on `email`, with no join needed. The decomposition is **dependency preserving**, as well as lossless.

Relation	**Minimal basis of projection**	**Enforced in `sql/ddl.sql` by**
`user`	g_1, g_2, g_3	`pk_user`, `uq_user_email`
`stock`	$g_4 - g_7$	`pk_stock`
`holds`	g_{14}, g_{15}	`pk_holds`
`transaction`	$g_8 - g_{13}$	`pk_transaction`